

Week4-Day-1-handson_Csharp_Gurunath_Rageni

Level-1 Problem 1: Student Grade Evaluator

Scenario

You are developing a console-based application in .NET 8 for a school. The application should evaluate a student's marks and assign a grade based on predefined rules.

Requirements

- Accept student name and marks (0-100).
- Use if-else statements to determine grade.
- Display grade as A, B, C, D or Fail.
- Handle invalid input using conditional checks.

Technical Constraints

- Use C# (.NET 8 Console Application).
- Use appropriate data types (string, int).
- Use if-else control flow.
- Do not use advanced concepts like classes or LINQ.

Sample Input

Enter Name: Rahul

Enter Marks: 78

Sample Output

Student: Rahul

Grade: B

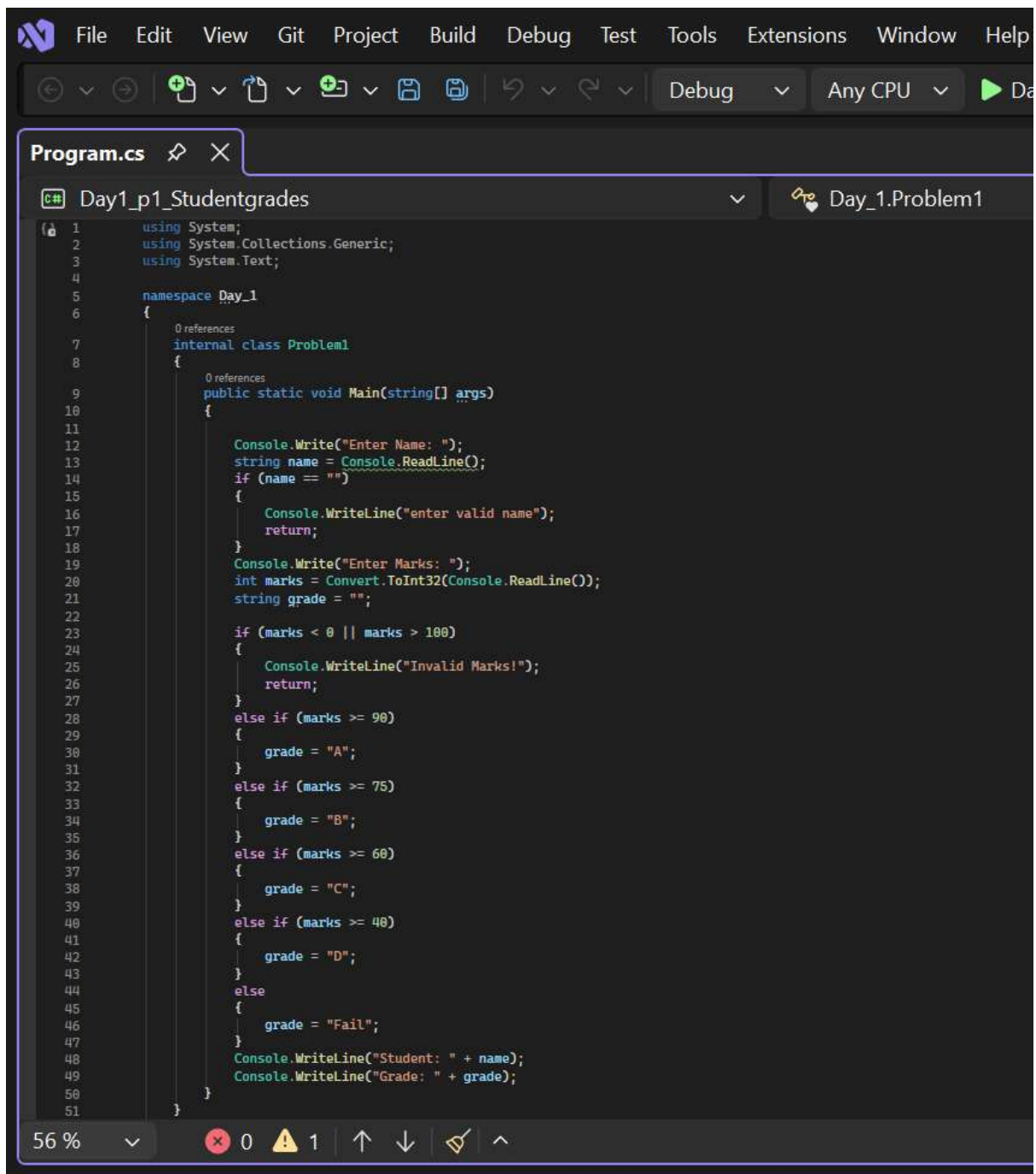
Expectations

Program should correctly evaluate grade and handle edge cases like marks below 0 or above 100.

Learning Outcome

Understand variables, data types, input/output handling and if-else control statements in C#.

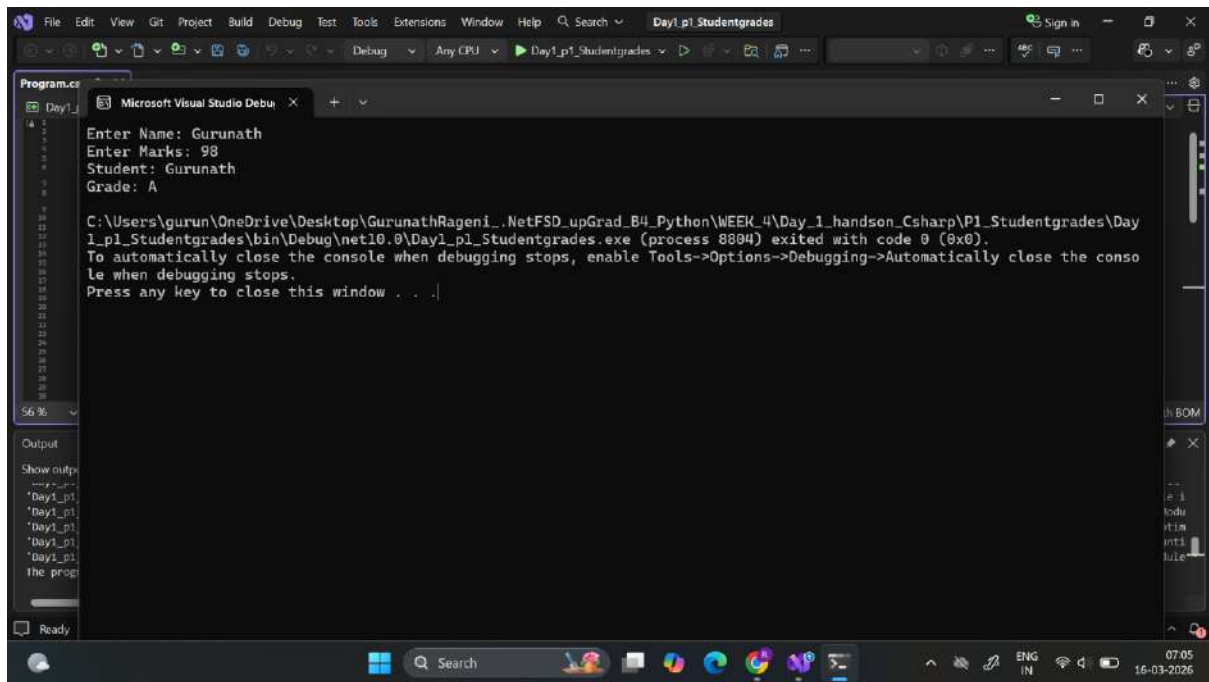
Code screenshot:



The image shows a screenshot of the Visual Studio IDE. The top menu bar includes File, Edit, View, Git, Project, Build, Debug, Test, Tools, Extensions, Window, and Help. Below the menu bar is a toolbar with icons for undo, redo, save, and other development actions. The main editor window displays a C# file named Program.cs. The code defines a namespace Day_1 containing an internal class Problem1. Inside Problem1, there is a public static void Main method that takes an array of strings as an argument. The Main method prompts the user to enter a name and marks, then calculates a grade based on the marks. The grade is determined by a series of if-else statements: marks < 0 or > 100 result in an invalid message; marks >= 90 result in grade 'A'; marks >= 75 result in grade 'B'; marks >= 60 result in grade 'C'; marks >= 40 result in grade 'D'; and marks < 40 result in grade 'Fail'. The program then prints the student's name and grade. The bottom status bar shows the zoom level at 56% and a warning icon with the number 1.

```
1 using System;
2 using System.Collections.Generic;
3 using System.Text;
4
5 namespace Day_1
6 {
7     internal class Problem1
8     {
9         public static void Main(string[] args)
10         {
11             Console.Write("Enter Name: ");
12             string name = Console.ReadLine();
13             if (name == "")
14             {
15                 Console.WriteLine("enter valid name");
16                 return;
17             }
18             Console.Write("Enter Marks: ");
19             int marks = Convert.ToInt32(Console.ReadLine());
20             string grade = "";
21
22             if (marks < 0 || marks > 100)
23             {
24                 Console.WriteLine("Invalid Marks!");
25                 return;
26             }
27             else if (marks >= 90)
28             {
29                 grade = "A";
30             }
31             else if (marks >= 75)
32             {
33                 grade = "B";
34             }
35             else if (marks >= 60)
36             {
37                 grade = "C";
38             }
39             else if (marks >= 40)
40             {
41                 grade = "D";
42             }
43             else
44             {
45                 grade = "Fail";
46             }
47             Console.WriteLine("Student: " + name);
48             Console.WriteLine("Grade: " + grade);
49         }
50     }
51 }
```

Output:



Level-1 Problem 2: Simple Calculator Using Switch

Scenario

Create a simple calculator application that performs basic arithmetic operations.

Requirements

- Accept two numbers from user.
- Accept operator (+, -, *, /).
- Use switch statement to perform operation.
- Display result.

Technical Constraints

- Use int or double data types.
- Use switch-case statement.
- Handle division by zero.

Sample Input

Enter First Number: 10
Enter Second Number: 5
Enter Operator: *

Sample Output

Result: 50

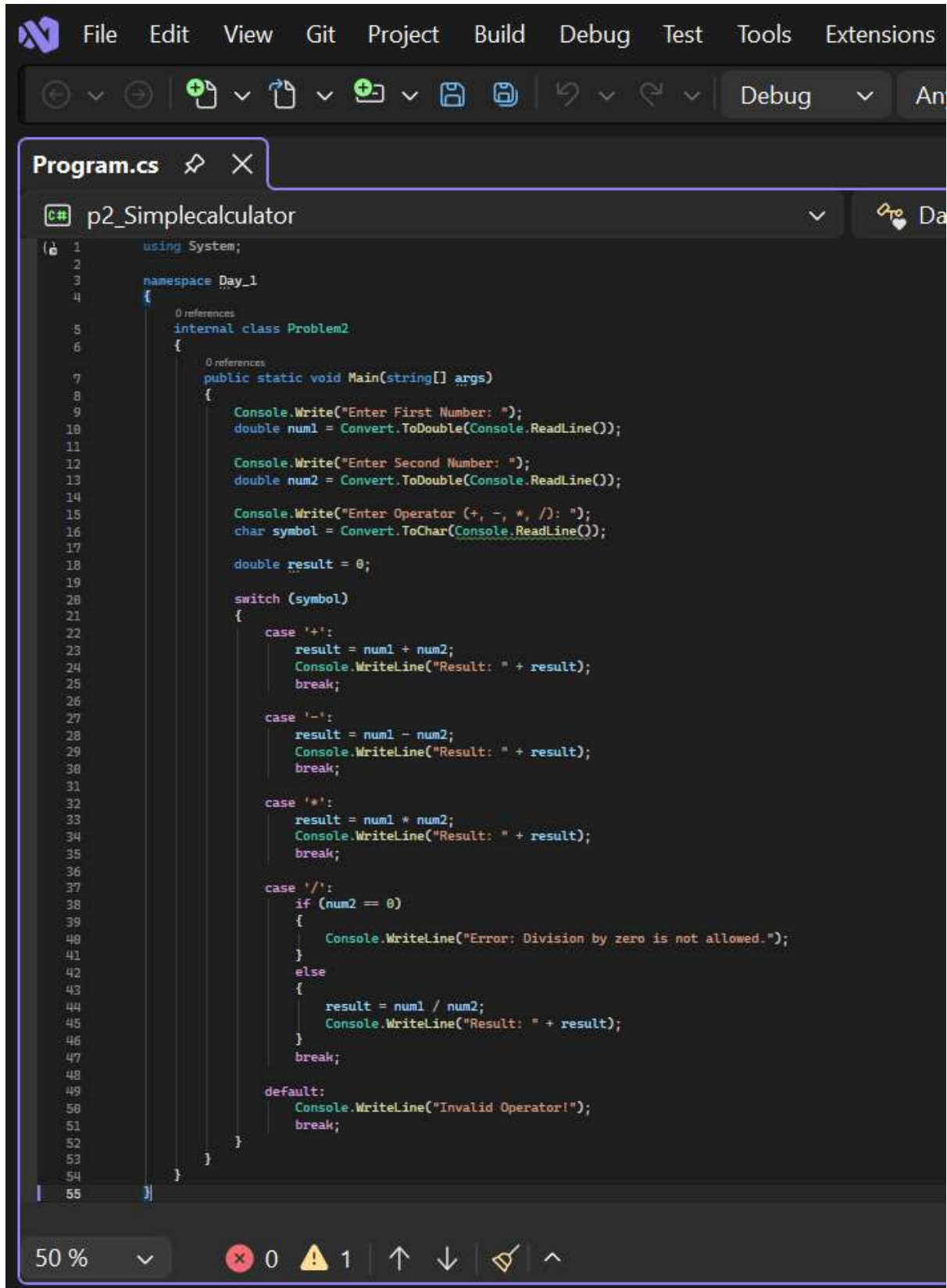
Expectations

Correct operator selection and proper validation of inputs.

Learning Outcome

Understand switch statements, arithmetic operators and control flow in C#.

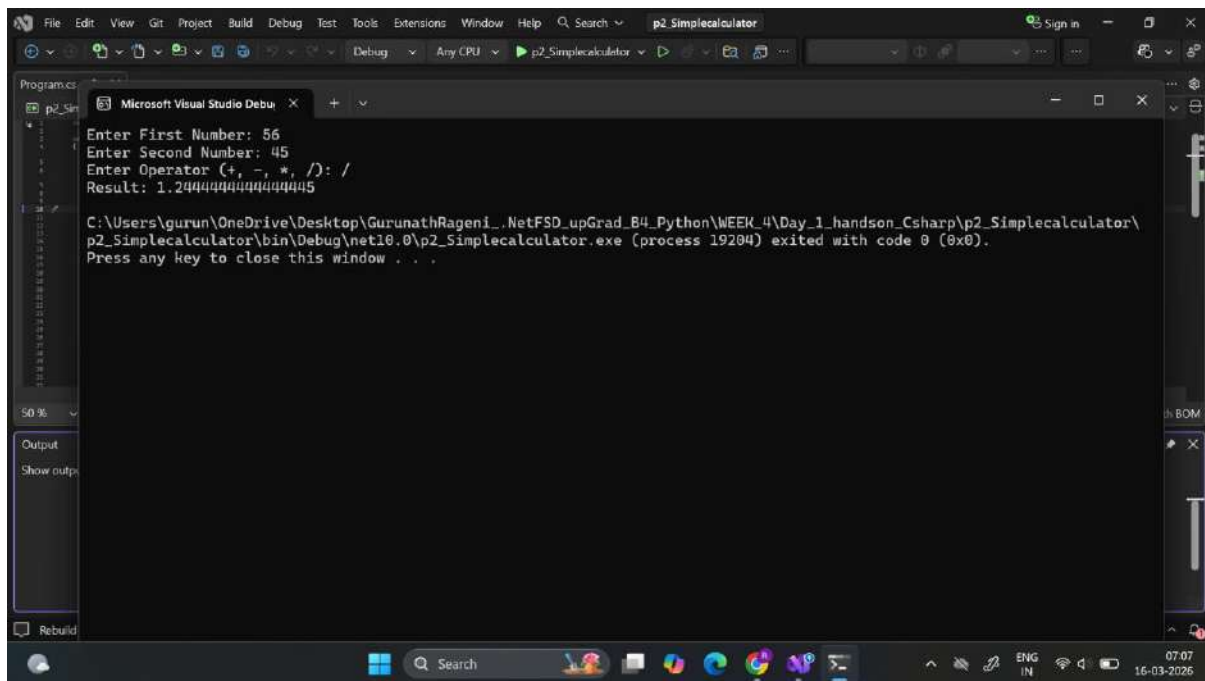
Code screenshot:



The screenshot shows the Visual Studio IDE with a C# file named `Program.cs` open. The code is for a simple calculator application. It uses the `System` namespace and defines an internal class `Problem2`. The `Main` method prompts the user for two numbers and an operator, then performs the calculation using a `switch` statement. The operators supported are `+`, `-`, `*`, and `/`. A `default` case handles invalid operators. The code also includes error handling for division by zero.

```
1 using System;
2
3 namespace Day_1
4 {
5     0 references
6     internal class Problem2
7     {
8         0 references
9         public static void Main(string[] args)
10         {
11             Console.Write("Enter First Number: ");
12             double num1 = Convert.ToDouble(Console.ReadLine());
13
14             Console.Write("Enter Second Number: ");
15             double num2 = Convert.ToDouble(Console.ReadLine());
16
17             Console.Write("Enter Operator (+, -, *, /): ");
18             char symbol = Convert.ToChar(Console.ReadLine());
19
20             double result = 0;
21
22             switch (symbol)
23             {
24                 case '+':
25                     result = num1 + num2;
26                     Console.WriteLine("Result: " + result);
27                     break;
28
29                 case '-':
30                     result = num1 - num2;
31                     Console.WriteLine("Result: " + result);
32                     break;
33
34                 case '*':
35                     result = num1 * num2;
36                     Console.WriteLine("Result: " + result);
37                     break;
38
39                 case '/':
40                     if (num2 == 0)
41                     {
42                         Console.WriteLine("Error: Division by zero is not allowed.");
43                     }
44                     else
45                     {
46                         result = num1 / num2;
47                         Console.WriteLine("Result: " + result);
48                     }
49                     break;
50
51                 default:
52                     Console.WriteLine("Invalid Operator!");
53                     break;
54             }
55         }
56     }
57 }
```

Output:



```
Enter First Number: 56
Enter Second Number: 45
Enter Operator (+, -, *, /): /
Result: 1.2444444444444445

C:\Users\gurun\OneDrive\Desktop\GurunathRageni_.NetFSD_upGrad_B4_Python\WEEK_4\Day_1_handson_Csharp\p2_Simplecalculator\
p2_Simplecalculator\bin\Debug\net10.0\p2_Simplecalculator.exe (process 19204) exited with code 0 (0x0).
Press any key to close this window . . .
```

Level-2 Problem 1: Employee Bonus Calculator

Scenario

Develop a console application that calculates employee bonus based on salary and years of experience.

Requirements

- Accept employee name, salary and years of experience.
- Use if-else and conditional operator.
- Bonus rules:
 - Experience < 2 years: 5% bonus
 - 2-5 years: 10% bonus
 - >5 years: 15% bonus
- Display final salary after bonus.

Technical Constraints

- Use double for salary.
- Use if-else and ternary operator.
- Use proper formatting for currency output.

Sample Input

Enter Name: Aisha
Enter Salary: 50000
Enter Experience: 4

Sample Output

Employee: Aisha
Bonus: 5000
Final Salary: 55000

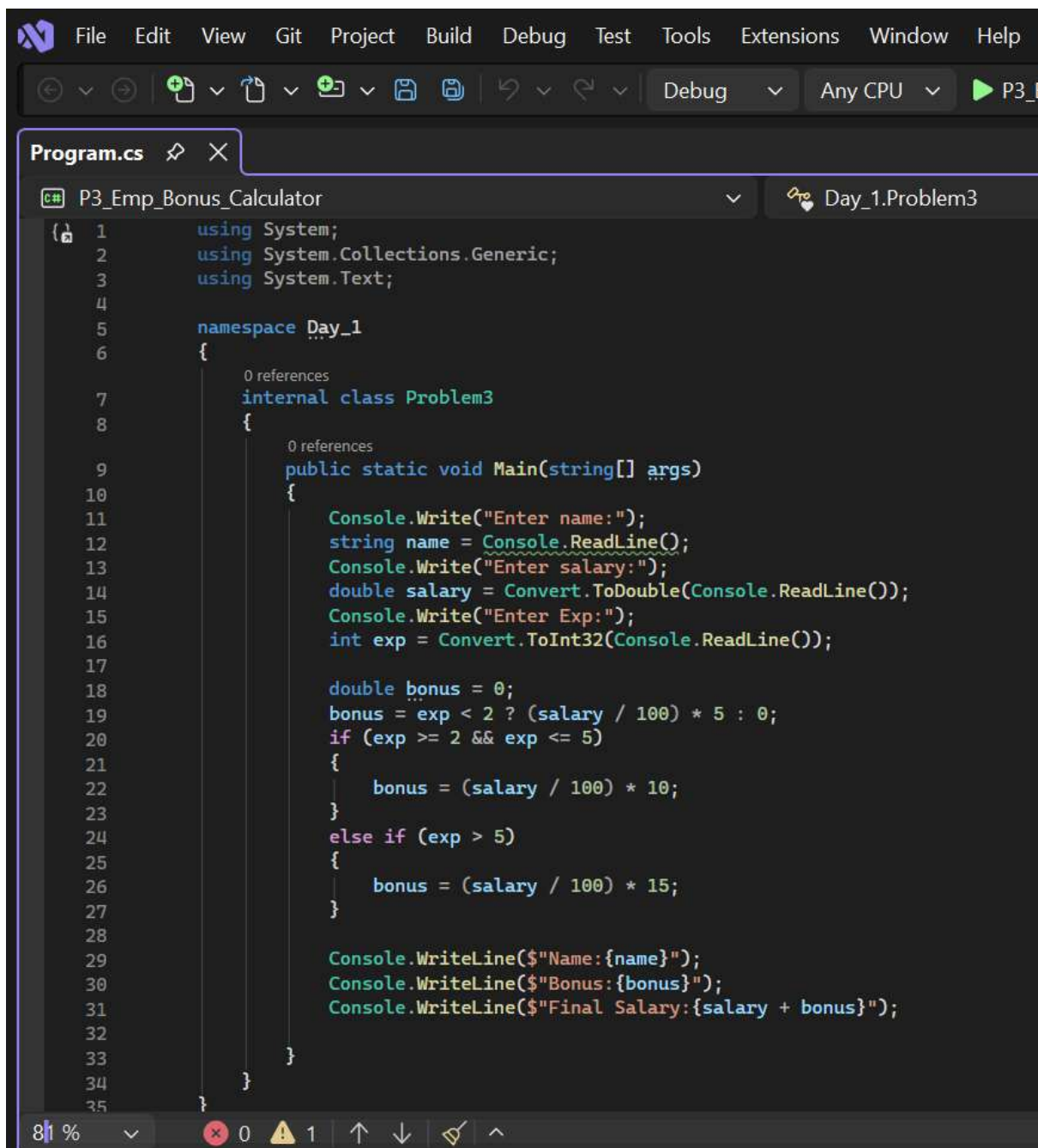
Expectations

Accurate bonus calculation and correct usage of control statements.

Learning Outcome

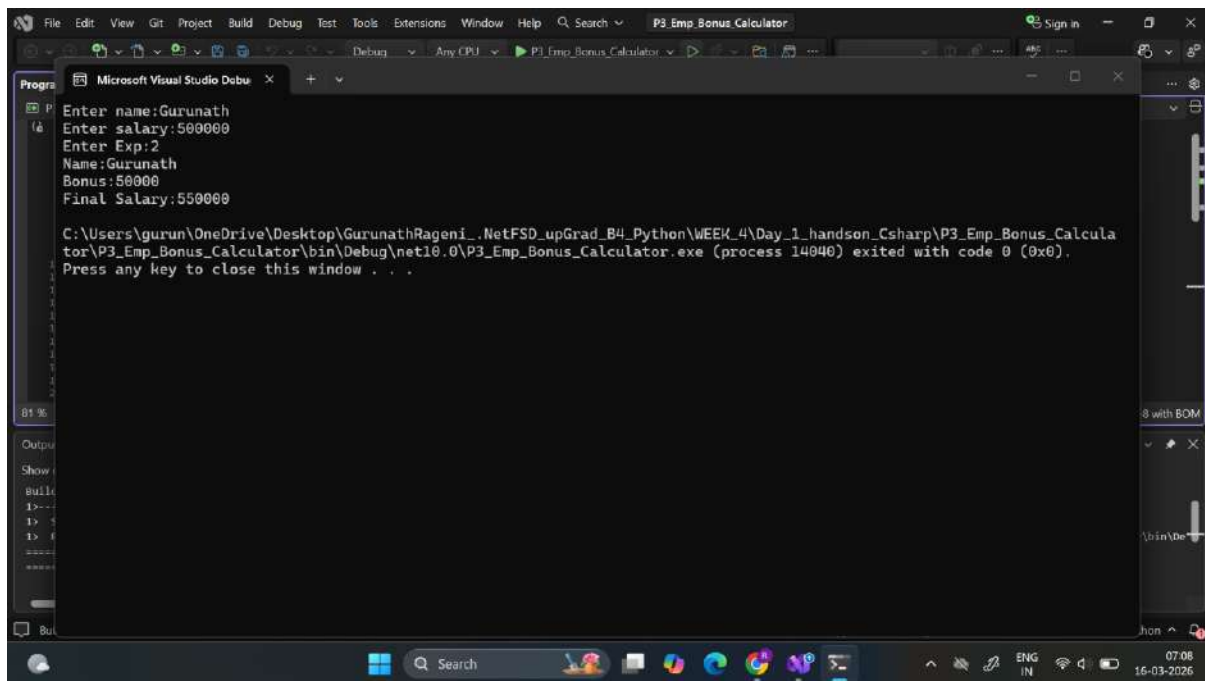
Apply conditional logic and arithmetic operations in real-world scenarios.

Code screenshot:



```
1  using System;
2  using System.Collections.Generic;
3  using System.Text;
4
5  namespace Day_1
6  {
7      0 references
8      internal class Problem3
9      {
10         0 references
11         public static void Main(string[] args)
12         {
13             Console.Write("Enter name:");
14             string name = Console.ReadLine();
15             Console.Write("Enter salary:");
16             double salary = Convert.ToDouble(Console.ReadLine());
17             Console.Write("Enter Exp:");
18             int exp = Convert.ToInt32(Console.ReadLine());
19
20             double bonus = 0;
21             bonus = exp < 2 ? (salary / 100) * 5 : 0;
22             if (exp >= 2 && exp <= 5)
23             {
24                 bonus = (salary / 100) * 10;
25             }
26             else if (exp > 5)
27             {
28                 bonus = (salary / 100) * 15;
29             }
30
31             Console.WriteLine($"Name:{name}");
32             Console.WriteLine($"Bonus:{bonus}");
33             Console.WriteLine($"Final Salary:{salary + bonus}");
34         }
35     }
```

Output:



```
Enter name:Gurunath
Enter salary:500000
Enter Exp:2
Name:Gurunath
Bonus:50000
Final Salary:550000

C:\Users\gurun\OneDrive\Desktop\GurunathRageni_.NetFSD_upGrad_B4_Python\WEEK_4\Day_1_handson_Csharp\P3_Emp_Bonus_Calculator\P3_Emp_Bonus_Calculator\bin\Debug\net10.0\P3_Emp_Bonus_Calculator.exe (process 14040) exited with code 0 (0x0).
Press any key to close this window . . .
```

Level-2 Problem 2: Number Analysis Using Loops

Scenario

Create a .NET 8 console application that analyzes numbers between 1 and N.

Requirements

- Accept a number N from user.
- Use loops to:
 - Count even numbers
 - Count odd numbers
 - Calculate sum of all numbers
- Display results.

Technical Constraints

- Use for or while loop.
- Use int data type.
- Avoid using arrays or collections.

Sample Input

Enter Number: 10

Sample Output

Even Count: 5

Odd Count: 5

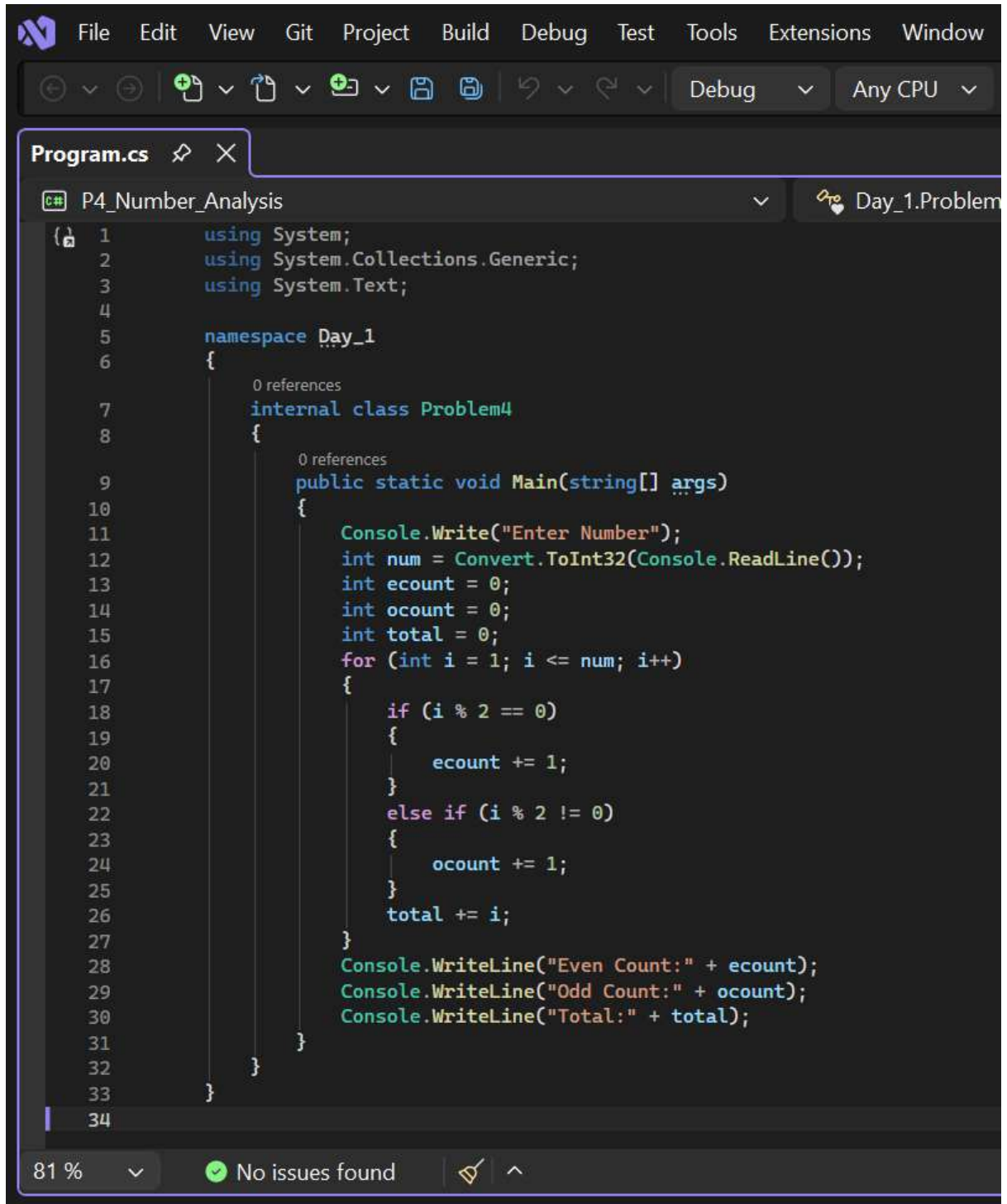
Sum: 55

Expectations

Proper loop usage and correct counting logic.

Learning Outcome

Strengthen understanding of loops, counters and accumulators in C#.

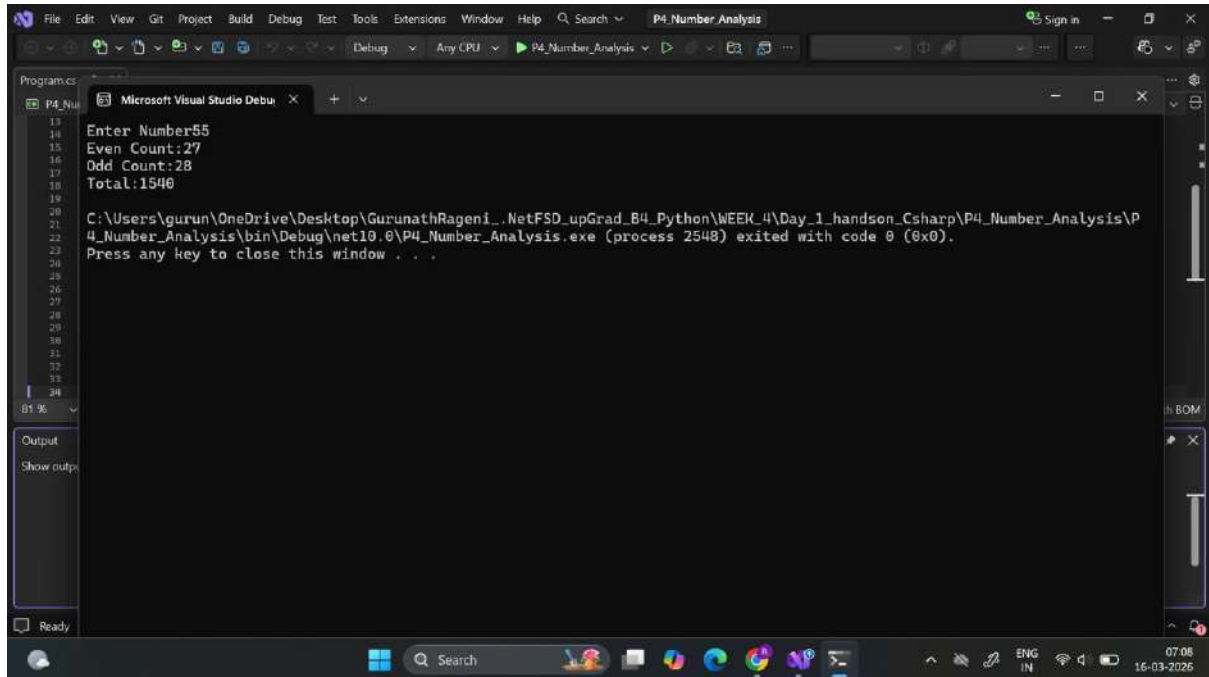


The screenshot shows the Visual Studio IDE with a C# file named Program.cs. The code is part of a project named P4_Number_Analysis and is within the namespace Day_1. It defines an internal class Problem4 with a public static Main method. The Main method prompts the user to enter a number, reads it, and then uses a for loop to calculate the count of even and odd numbers, as well as the total sum of the numbers from 1 to the entered number. The results are displayed on the console.

```
1 using System;
2 using System.Collections.Generic;
3 using System.Text;
4
5 namespace Day_1
6 {
7     0 references
8     internal class Problem4
9     {
10        0 references
11        public static void Main(string[] args)
12        {
13            Console.Write("Enter Number");
14            int num = Convert.ToInt32(Console.ReadLine());
15            int ecount = 0;
16            int ocount = 0;
17            int total = 0;
18            for (int i = 1; i <= num; i++)
19            {
20                if (i % 2 == 0)
21                {
22                    ecount += 1;
23                }
24                else if (i % 2 != 0)
25                {
26                    ocount += 1;
27                }
28                total += i;
29            }
30            Console.WriteLine("Even Count:" + ecount);
31            Console.WriteLine("Odd Count:" + ocount);
32            Console.WriteLine("Total:" + total);
33        }
34    }
35 }
```

81 % No issues found

Output:



```
13 Enter Number:55
14 Even Count:27
15 Odd Count:28
16 Total:1546
17
18 C:\Users\gurun\OneDrive\Desktop\GurunathRageni_.NetFSD_upGrad_B4_Python\WEEK_4\Day_1_handson_Csharp\P4_Number_Analysis\P
19 4_Number_Analysis\bin\Debug\net10.0\P4_Number_Analysis.exe (process 2548) exited with code 0 (0x0).
20 Press any key to close this window . . .
21
22
23
24
25
26
27
28
29
30
31
32
33
34
```

Output
Show output

Ready

07:08
16-03-2026